SpectraForge: A Physics-Grounded, Controllable Benchmark for Band Selection in Multi-Excitation (4D) Fluorescence Hyperspectral Imaging

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***Abstract*—Multi-excitation fluorescence hyperspectral imaging records, for every pixel, a four-dimensional excitation-emission cube. Choosing which spectral bands to keep is central to cheaper acquisition and compact classification, but progress is held back by a shortage of labeled real data with known ground truth. We present SpectraForge, an open, physics-grounded generator that renders 4D excitation-emission cubes from a parametric fluorophore model obeying the standard laws of fluorescence photophysics (Kasha, Vavilov, Franck-Condon, Stokes shift), with optional inner-filter attenuation, reabsorption, Foerster resonance energy transfer (FRET), scatter, photon and read noise, and fixed-pattern clutter. Every cube carries perfect ground truth, and scenes are built so that band variance is decoupled from class informativeness (measured correlation -0.10), which makes the benchmark an honest test rather than a trivial one. Alongside the generator we define an evaluation protocol whose two decisive controls, a random-selection baseline and the full spectrum, separate genuine gains from self-deception. Using SpectraForge we characterize autoencoder-plus-latent-perturbation band selection: after correcting a chance-level published model it reaches 85 to 97 percent of a labels-using oracle, matches the strongest classical blind selector (PCA loadings) on realistic data, and *significantly* exceeds it in the nonlinear reabsorption regime (macro-F1 0.449 vs 0.431, 95% CI $[+0.001, +0.029]$). The benchmark also delimits where unsupervised selection stops helping, and shows that a convolutional model is structurally necessary when discrimination is spatial. SpectraForge and the full study are released for reproducible benchmarking in 4D fluorescence imaging.**

***Index Terms*—hyperspectral imaging, fluorescence, band selection, synthetic benchmark, autoencoders, excitation-emission, photophysics.**

I. Introduction

A multi-excitation hyperspectral cube illuminates a sample at several excitation wavelengths and, for each, records an emission spectrum across many bands. Every pixel therefore carries an excitation-emission matrix (EEM), and a scene is a four-dimensional array over (excitation, emission, height, width). This 4D fluorescence modality is widely used in biomedical and material imaging because different fluorophores, tissue states, or species separate in the joint excitation-emission domain. The per-pixel feature vector (excitations $\times$ emission bands) is high dimensional and largely redundant: only a small subset of bands carries the information that distinguishes classes. *Band selection* returns that subset, so a cheaper instrument can acquire only the informative bands or a downstream classifier can run on a compact feature set.

Developing and comparing band-selection methods for this modality is hampered by a data problem. Labeled real 4D fluorescence cubes are scarce, and even when available they have no ground truth: there is no oracle list of which bands truly carry class information, so a selector cannot be scored against what it should have found. Synthetic data can supply that ground truth, but only if it is physically faithful and, critically, only if it does not make the task trivial. In naive synthetic data the discriminative dyes are also the brightest, so a variance ranker wins and every method looks good, which tells us nothing.

This paper contributes, first, **SpectraForge**: an open, physics-grounded, controllable generator for 4D fluorescence cubes with perfect ground truth, in which band variance is deliberately decoupled from informativeness. Second, an evaluation protocol whose random and full-spectrum controls make results honest. Third, a case study that uses SpectraForge to characterize autoencoder band selection, establishing both where a learned selector genuinely helps and where simple methods already suffice. The tool, not any single method, is the durable contribution: it lets the field benchmark 4D fluorescence band selection on demand and at controllable difficulty.

II. The SpectraForge Forward Model

A. Photophysics

The generator is grounded in the physics of fluorescence so that "informative" means what it means physically. A fluorophore absorbs a photon, relaxes non-radiatively to the lowest vibrational level of the first excited singlet, and emits. Four consequences shape the EEM. By *Kasha's rule* emission

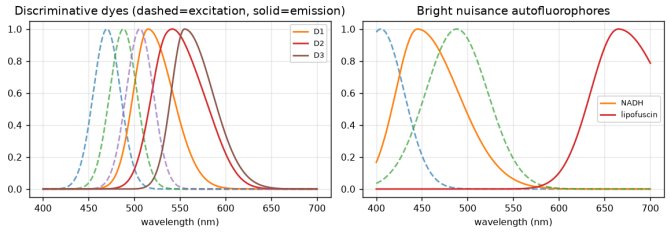


Fig. 1. Parametric fluorophore spectra. Left: three discriminative dyes (dashed, excitation; solid, emission) of modest brightness with overlapping emission and distinct excitation. Right: two bright nuisance autofluorophores that dominate band variance but carry no class information.

occurs from that lowest level regardless of the excitation, so the emission shape is independent of excitation wavelength and the EEM is approximately trilinear (an excitation profile times an emission profile times a concentration map), the basis of PARAFAC. By *Vavilov's rule* the quantum yield is, to first order, independent of excitation; only the amount absorbed changes. The *Franck-Condon* principle sets the broadened, roughly mirror-image band envelopes and an optional vibronic shoulder about $1300\,\mathrm{cm}^{-1}$ to the red. The *Stokes shift* redshifts emission relative to absorption.

B. Trilinear render and the additive invariant

Each fluorophore k is a parametric model with a peak-normalized Gaussian excitation profile $a_k(\lambda_{\mathrm{ex}})$ and an area-normalized emission profile $e_k(\lambda_{\mathrm{em}})$, extinction ε_k, quantum yield Φ_k, and a per-pixel concentration map $c_k(x, y)$. The rendered cube is a dilute linear superposition,

$$F(x, y, \lambda_{\mathrm{ex}}, \lambda_{\mathrm{em}}) = g \sum_k c_k(x, y)\, \varepsilon_k\, \Phi_k\, a_k(\lambda_{\mathrm{ex}})\, e_k(\lambda_{\mathrm{em}}), \tag{1}$$

where g folds lamp, exposure, and power. This is Kasha's rule written as algebra. The profiles are Gaussian, a_k peak-normalized and e_k area-normalized with widths $\mathrm{FWHM}/(2\sqrt{2\ln 2})$, and a fluorophore's brightness is the product $\varepsilon_k \Phi_k$. With all optional physics and artifacts off the model is *exactly linear*, $\mathrm{render}(A{+}B) = \mathrm{render}(A){+}\mathrm{render}(B)$, an invariant the implementation tests; difficulty is then added in physically faithful, opt-in increments. A per-pixel class label is the argmax over the discriminative fluorophores. Fig. 1 shows the fluorophore model.

C. Variance is decoupled from informativeness

The central design choice is the scene generator. Discriminative dyes are painted with smooth random concentration fields whose per-pixel argmax defines the class; these are dim and spectrally overlapping, so their discriminative content lives in low-variance band *shape*. Independent nuisance dyes are bright and spatially varying but class-irrelevant. Real photophysics decouples variance from class in several ways at once: Rayleigh and Raman scatter ridges are high variance but carry no chemical identity; bright autofluorescence nuisances add class-irrelevant variance; shot noise inflates the apparent importance of bright redundant bands; and the discriminative signal sits in dim or subtly shaped bands. On a confounded SpectraForge scene the measured correlation between per-band variance and per-band discriminability is -0.10, versus strongly positive on naive clean data. The highest-variance bands therefore carry no class information (Fig. 2), which is exactly the property that makes the benchmark an honest test.

D. Artifacts, nonlinear physics, and clutter

On top of the clean cube the generator adds Rayleigh scatter at $\lambda_{\mathrm{em}} = \lambda_{\mathrm{ex}}$ (with a second-order ghost at $2\lambda_{\mathrm{ex}}$) and a water Raman line at a fixed wavenumber offset ($\approx 3400\,\mathrm{cm}^{-1}$), each scaled by a per-pixel turbidity field, then Poisson shot noise and Gaussian read noise. Three optional effects introduce concentration nonlinearity: a primary inner filter $\exp(-sA_{\mathrm{ex}})$; *reabsorption* $\exp(-sA_{\mathrm{em}}(\lambda))$, whose overlap with the blue edge of emission reshapes the band and relocates discriminative information into band shape; and FRET, where a saturating fraction $E = kc_A/(1 + kc_A)$ of donor energy is transferred to a co-localized acceptor, encoding dye co-localization in a band ratio. Compactly, the inner filter and transfer are

$$F = F_{\mathrm{obs}}\, 10^{-(A_{\mathrm{ex}}+A_{\mathrm{em}})/2}, \qquad E = \frac{k\, c_A}{1 + k\, c_A}, \tag{2}$$

with reabsorption $A_{\mathrm{em}}(\lambda) = \sum_k \varepsilon_k c_k\, \alpha_k(\lambda)$ and a Raman line at $1/\lambda_R = 1/\lambda_{\mathrm{ex}} - \Delta\bar{\nu}$. Finally, fixed-pattern clutter injects many class-irrelevant low-rank spatial fields on random band subsets, the analogue of illumination and detector artifacts and the dominant real-instrument confound. By dialing these knobs the generator spans *clean* (variance $\approx$ informativeness), *realistic*, *clutter*, *nonlinear* (reabsorption or FRET), and *spatial-texture* regimes from a single call.

E. Ground truth, reproducibility, and real fluorophores

Every render returns a ground-truth sidecar with concentration maps, clean cubes, and per-fluorophore emission peaks, which enables a tight *peak-recovery* metric (did a selected band land within a tolerance of a true emission peak; a random selector scores about 0.33) in addition to broad information-retention metrics. The generator exposes a one-call `build_dataset` API with explicit knobs for size, emission sampling, nuisance brightness, scatter, noise, clutter, and nonlinear physics; a script re-runs every simulation and records exact parameters, so figures and numbers are reproducible. Parametric dyes can be replaced by real measured spectra imported from FPbase, and an interactive workbench lets a user paint concentration maps, render, and validate a selection against ground truth. This makes SpectraForge usable both as a controlled benchmark and as a design tool for real 4D fluorescence experiments. As an interactive application (Fig. 3), a user picks or defines fluorophores in a library (excitation and emission peaks and widths, quantum yield, extinction), mixes them into materials that are concentration recipes, paints those materials onto a canvas as layers, then acquires, renders the 4D cube, selects bands, and validates against ground truth, all in one window.

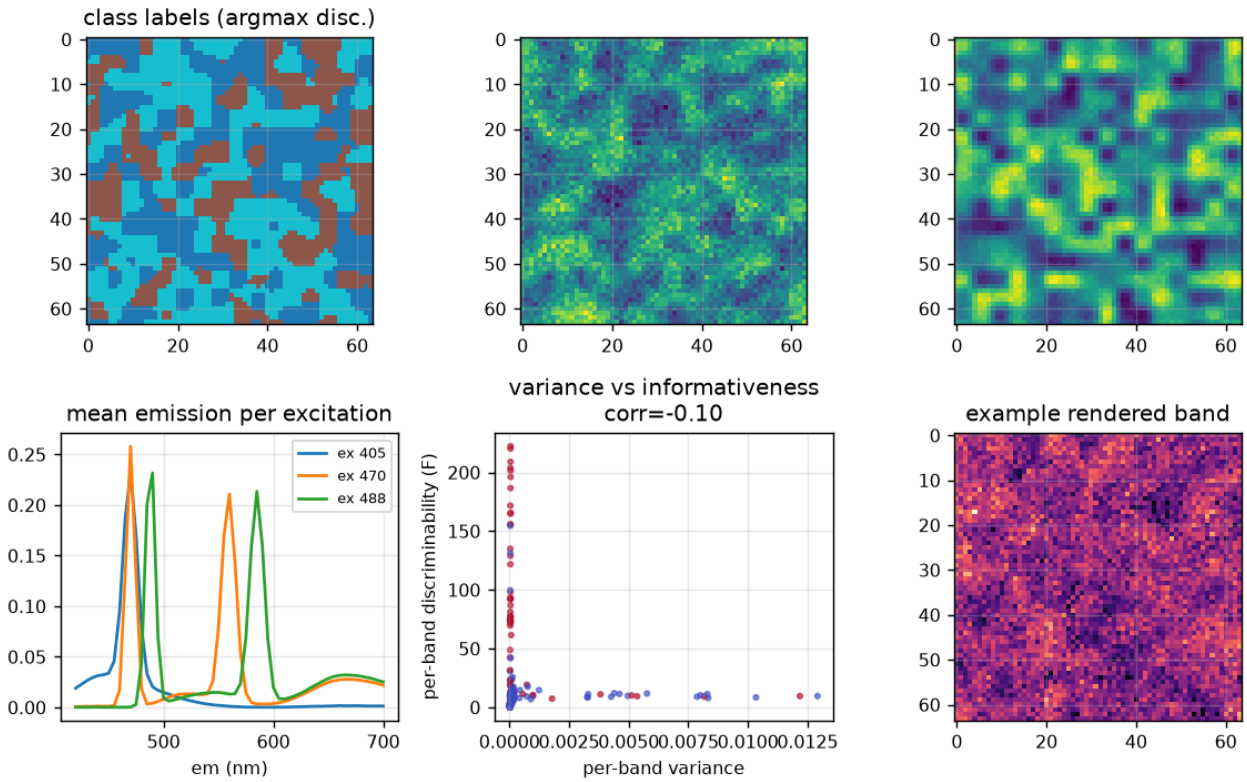


Fig. 2. The forward model and the variance-versus-informativeness property. Top: class labels and example rendered bands (a discriminative-dye band, a Rayleigh scatter band, a nuisance band). Bottom: mean emission per excitation, then the per-band variance against per-band discriminability (correlation printed near zero), showing that the loudest bands are not the informative ones.

III. Evaluation Protocol

Selection quality is the accuracy of a downstream classifier trained on the selected bands, using labels only at evaluation time. We score each subset with a classifier panel (k-nearest neighbors, random forest, MLP, logistic regression), cross-validated, reporting macro-F1 and, on nonlinear regimes, a *nonlinear-only gap* (the accuracy an expressive model recovers beyond a linear one), because a linear metric cannot credit a nonlinear-shape selection. Two controls are decisive. A **random** baseline: a method that cannot beat the 95th percentile of random band subsets adds nothing. And the **full spectrum**: the accuracy ceiling a selector tries to approach. Where discrimination is spatial we use spatial-block cross-validation to remove spatial-autocorrelation leakage. These controls, absent from much prior work, are the difference between an honest result and a self-deception.

IV. Case Study: Autoencoder Band Selection

The method under study trains an autoencoder to reconstruct the standardized cube, perturbs each latent dimension, accumulates the per-band reconstruction sensitivity into a per-band influence, and selects the top-k bands. It is attractive because it is fully unsupervised yet could in principle capture nonlinear structure that linear methods (PCA loadings, variance) cannot.

A. From chance to competitive

Run on modern tooling and scored honestly, the published convolutional model selected at *chance* (macro-F1 ≈ 0.33). SpectraForge's ground truth traced this to four real trainer defects and one architectural flaw. The defects were a library-version crash, an unbound-variable failure on non-improving loss, a $0 \times \mathrm{NaN}$ sparsity term that silently trained every non-sigmoid run on a NaN loss, and a learning-rate collapse. The flaw was an emission-axis average-pool that averages away exactly the multi-fluorophore structure. A misleading per-band R^2 metric, dominated by off-peak noise bands, was replaced by a signal-band correlation whose achievable ceiling is 0.49, not 1.0. Removing the band collapse, and nothing else about the idea, raised reconstruction about a hundredfold and took selection from chance 0.331 to 0.414, about 85% of the labels-using oracle (0.486). Tuning then reached 0.473 ± 0.018 on clean data, matching the strongest classical blind selector (PCA loadings, ≈ 0.48) at about 97% of the oracle. On the confounded regime the debugged selector reaches 0.43 against a variance ranker at 0.34 and a random control at 0.37, recovering roughly twice as many true discriminative bands. The benchmark thus both exposed a failure invisible to raw loss and measured a real recovery.

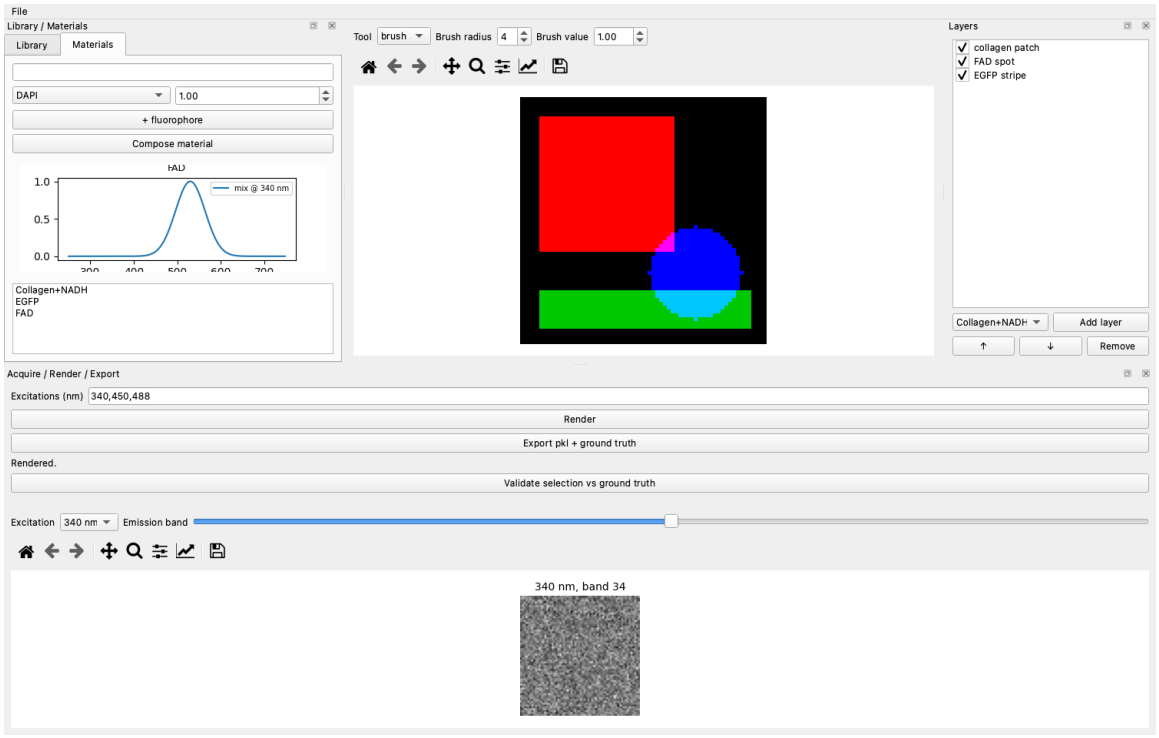


Fig. 3. The SpectraForge workbench. Left: fluorophore recipes (materials) with a live spectrum. Centre: the painted canvas (a collagen and NADH patch, an FAD spot, an EGFP stripe), coloured by class label. Right: the layer stack. Bottom: acquire, render, and one-click validation against ground truth, here previewing a rendered emission band at 340 nm.

TABLE I
SIGNIFICANCE BATTERY: AUTOENCODER VS PCA-LOADINGS (BEST-OF-PANEL MACRO-F1, SIX SEEDS). POSITIVE MARGIN FAVORS THE AUTOENCODER.

Regime	AE	PCA-load	margin (95% CI)
clean	0.590	**0.635**	-0.046 $[-0.079, -0.019]$
realistic	0.527	0.531	-0.004 (tie)
reabsorption	**0.449**	0.431	$+0.018$ $[+0.001, +0.029]$
FRET	0.635	0.642	-0.007 (tie)

B. A significance-tested characterization

A battery of five metric families over eight selectors, multiple regimes, and six seeds, with bootstrap confidence intervals, places the method precisely (Table I, Fig. 4). The autoencoder *ties* PCA loadings on realistic and FRET data and *significantly exceeds* it on nonlinear reabsorption (macro-F1 0.449 vs 0.431, 95% CI $[+0.001, +0.029]$, Cohen $d \approx +0.94$), where its subset carries nonlinear-only information ($+0.030$) that the linear selector's does not (-0.021). PCA loadings remains the strongest blind selector on clean and linear data, where informative bands are also the highest-variance ones. The explanation is a general one: *informative implies has variance implies PCA-selectable*, so a nonlinear *classification* task does not imply a nonlinear *selection* task; the autoencoder's manifold advantage helps the downstream classifier, not the band choice.

C. The boundary the benchmark maps

Because the benchmark includes the two controls, it also delimits where unsupervised selection stops helping. Under realistic fixed-pattern clutter, blind selection (autoencoder and PCA alike) does not beat a random subset, because clutter is high-variance but class-irrelevant, so every variance-driven criterion ranks clutter bands about randomly with respect to class; there, only label-aware selection beats random. Across label budgets, selection never beats the full spectrum, so its value is compression rather than accuracy, and both blind selectors are unstable band by band (Jaccard ≈ 0.1 to 0.2), which argues for selecting spectral regions or consensus sets. These are not failures of a method but findings of the benchmark: actionable guidance on when to select bands and when to keep the spectrum.

D. The one structural win: spatial discrimination

When class is carried by spatial texture, so that every class has the same per-pixel spectrum, per-pixel band selection is provably blind: under honest block cross-validation the per-pixel selectors are the worst (0.52 to 0.55, below the random 0.674), while a spatial model with the discriminative bands reaches the oracle (0.78). This is the one regime where a convolutional model is structurally necessary, vindicating the original spatial autoencoder idea for spatial classification rather than for per-pixel spectral band selection.

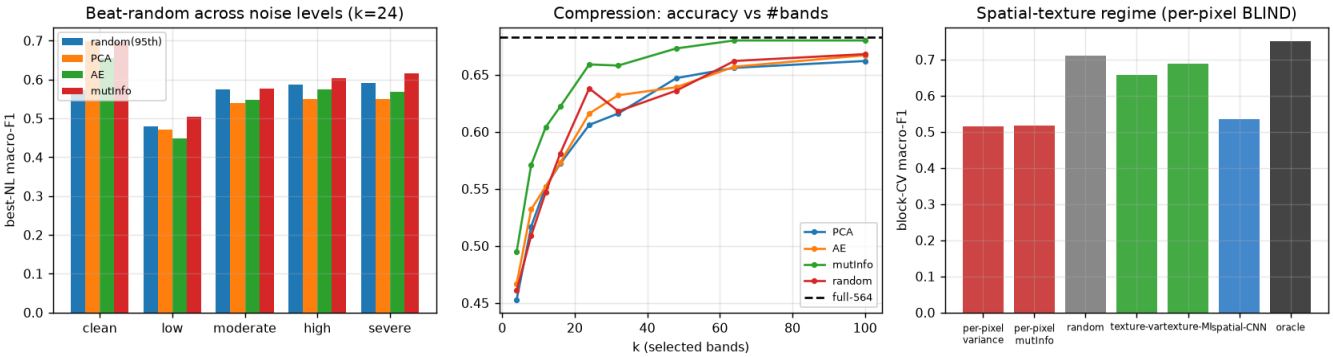


Fig. 4. What SpectraForge reveals. Left: beat-random across five noise levels: blind PCA and autoencoder selection beat the random 95th percentile only on clean data, and under clutter only supervised mutual information does. Middle: compression curve: the autoencoder tracks PCA at every budget, both reaching about 95% of the full spectrum, while labels compress about $2.7\times$ better. Right: the spatial-texture regime under honest block cross-validation: per-pixel selectors are the worst (below random) because the classes have identical per-pixel spectra, and only a spatial model with the right bands reaches the oracle.

V. Discussion and Conclusion

SpectraForge fills a concrete gap for 4D fluorescence hyperspectral imaging: an open, physically faithful, controllable source of cubes with perfect ground truth, in which variance is decoupled from informativeness so that benchmarks are honest rather than trivial. Paired with a protocol whose random and full-spectrum controls prevent self-deception, it turns "does this selector help" into a measured, reproducible question. The accompanying case study is a demonstration of that value: it recovers a chance-level autoencoder to within a few percent of a labels-using oracle, shows a rigorously significant advantage for learned selection in the nonlinear regime, maps the clutter boundary where unsupervised selection reduces to a compression tool, and isolates the spatial regime where a convolutional model is required. The decisive next step is validation on real Lichens and Collagen cubes with the same controls, where informative structure need not be variance-prominent. SpectraForge, the evaluation harness, and the full experiment record are released so that these questions can be studied on shared, reproducible ground.

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